

From Pathology to Radiology: Evaluating the Applicability of Pathology Foundation Models

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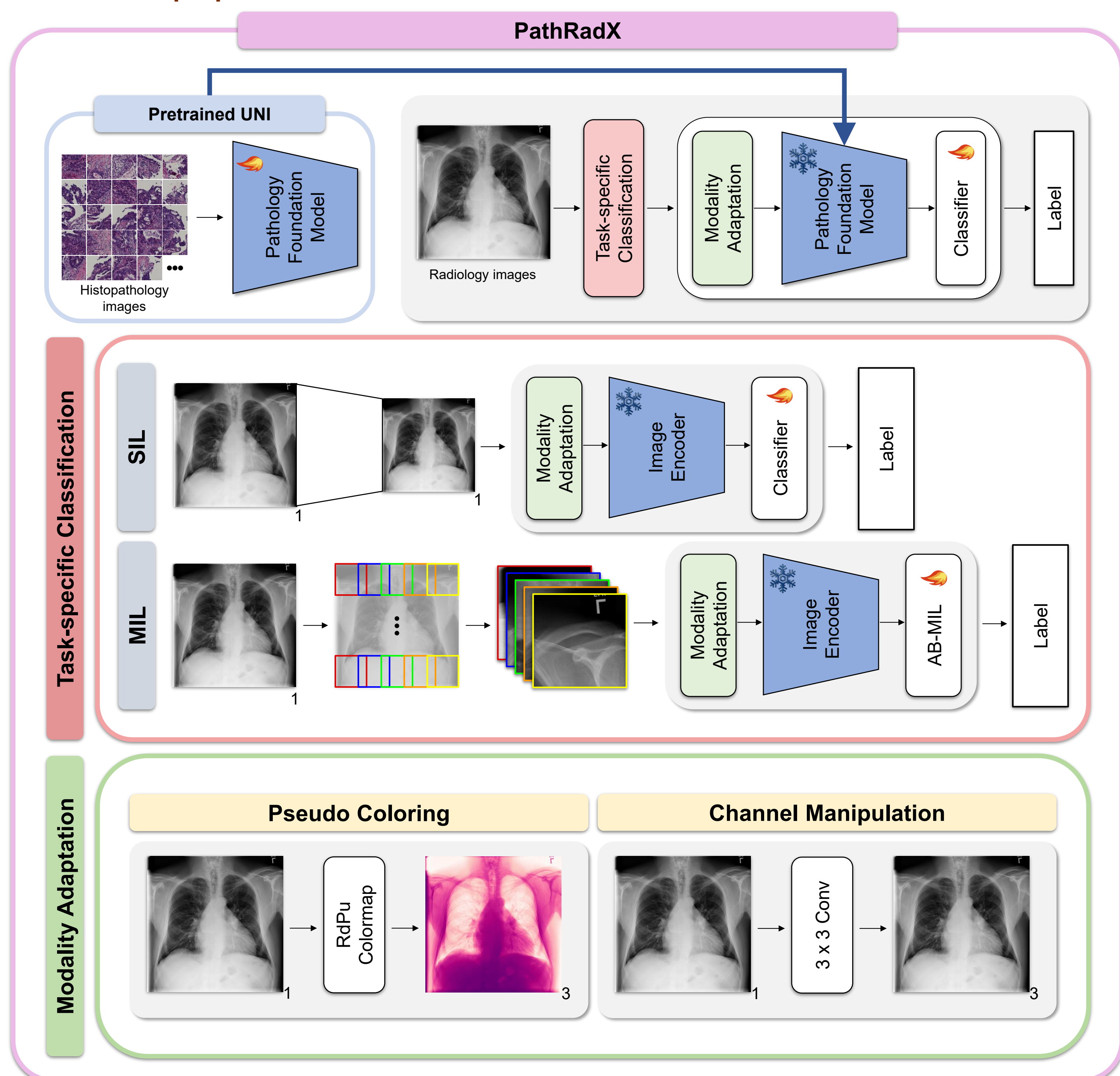
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INTRODUCTION

- Nowadays, with the recent advancements in deep learning and the increasing accessibility to large datasets, foundation models have gained considerable attention.
- As foundation models demonstrate strong adaptability across diverse tasks and datasets, several foundation models have been developed in medical domains.
- Yet, the adaptability and applicability of pathology foundation models to other medical imaging modalities remain largely unexplored.
- With the hypothesis that pathology foundation models possess sufficient generalization ability to be effectively adapted to other medical imaging modalities, such as radiology image, this study proposes PathRadX, a cross-modal framework that adapts a pathology foundation model to tasks specific to radiology.

METHODS

Overview of the proposed PathRadX framework



- Pathology Foundation Model**
 - Among various pathology foundation models, UNI was used as an image encoder of PathRadX. UNI is a self-supervised vision encoder for anatomic pathology based on the ViT architecture, pretrained on Mass-100K dataset using DINOv2.
- Task-specific Classification**
 - Single Instance Learning (SIL)** : Input radiology images were resized to 224 x 224 pixels to match the input spatial size of UNI.
 - Multiple Instance Learning (MIL)** : Multiple overlapping patches of size 224 x 224 pixels with an overlap of 80 pixels or 24 pixels were generated from original input images. Extracted features from each patch were then aggregated using attention-based MIL (AB-MIL).
- Modality Adaptation**
 - Strategies to apply gray-scaled radiology images to UNI, designed to handle three-channel RGB pathology images: Pseudo Coloring, Channel manipulation.
 - Pseudo coloring** : apply a pseudo-colorization technique using “RdPu” colormap from OpenCV.
 - Channel manipulation** : single 2D convolution layer to increase the number of channels while preserving the spatial size of input image.

Datasets

- MURA**
 - A large dataset of musculoskeletal radiographs containing 40,561 X-ray images from 14,863 studies, consisted of 9,045 normal and 5,818 abnormal radiographic studies.
 - RSNA Pneumonia Dataset (RSNA_p)**
 - Public dataset used during RSNA Pneumonia Detection Challenge held in 2018.
 - Containing 26,686 images of chest X-ray images.
 - MedFMC (MedFMC_c)**
 - Containing 22,349 images in total, covering 5 different modalities, X-ray, pathology, endoscopy, digital camera, and retinography.
 - Among them, 4,848 chest X-ray images with a spatial size of 1024 x 1024 pixels were used.
- The MURA dataset was split into training, validation, and test sets with ratios of 0.92, 0.037, and 0.042, respectively.
- Each of the RSNA_p and MedFMC_c datasets was split into training, validation, and test sets with ratios of 0.8, 0.1, and 0.1, respectively.

Comparative Models for the PathRadX framework

- To examine the potential of the PathRadX, several radiology foundation models were used as a comparison.
- SAM-Med2D**
 - built by fine-tuning SAM with approximately 4.6M medical images and 19.7M corresponding masks.
 - MedCLIP**
 - A vision-language model designed for medical image analysis, pretrained on a large dataset of chest X-ray images paired with clinical text, such as CheXpert and MIMIC-CXR.
 - MedFMC (MedFMC_c)**
 - Utilize PubMedBERT as a text encoder and ViT model as an image encoder.
 - Pre-trained on a large biomedical dataset, PMC-15M.

Network Training

- Binary Cross Entropy Loss (BCE Loss)

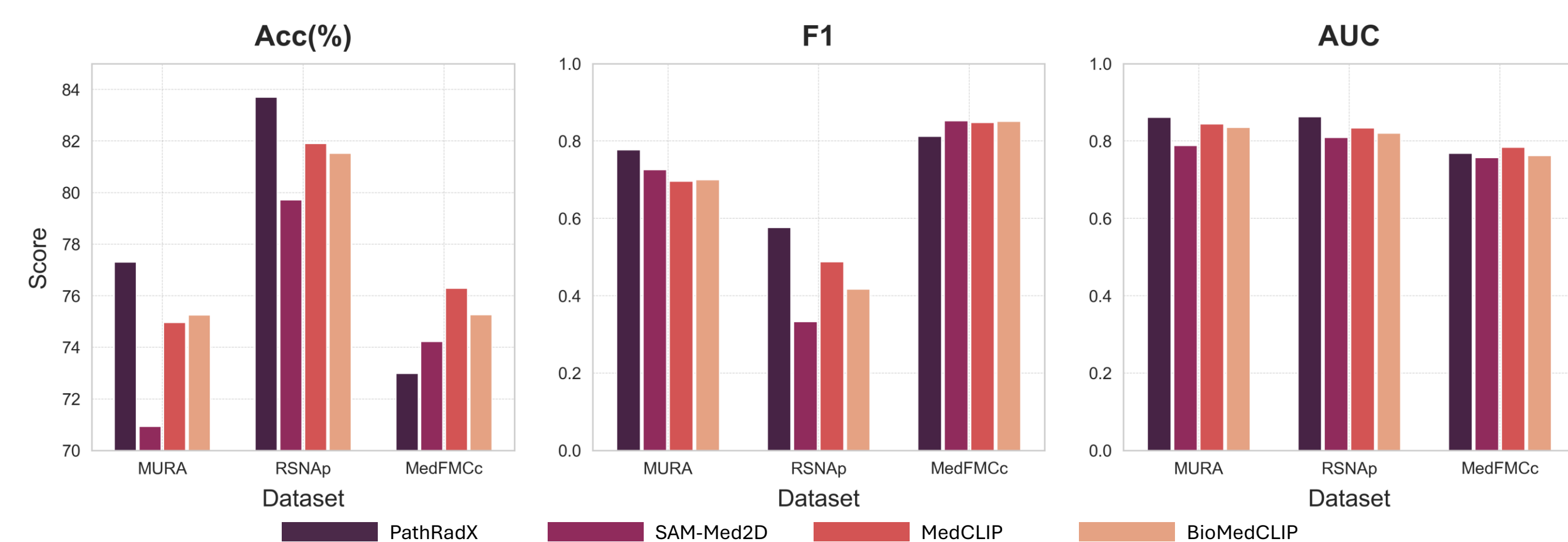
$$L_{BCE}(y, p) = -\frac{1}{N} \sum_{i=1}^N \{y_i \log(p(y_i)) + (1 - y_i) \log(1 - p(y_i))\}$$

- BCE loss is widely used for binary classification task to measure the difference between the predicted probability and the ground truth label.
- Except parameters of image encoders of each model, all parameters were updated during network training.
- All models were trained for 100 epochs using Adam optimizer, 0 weight decay and beta values (β_1 , β_2) set to 0.9 and 0.999, with OneCycleLR scheduler (initial learning rate of 0.001).

RESULTS

Comparison of classification performance across models.

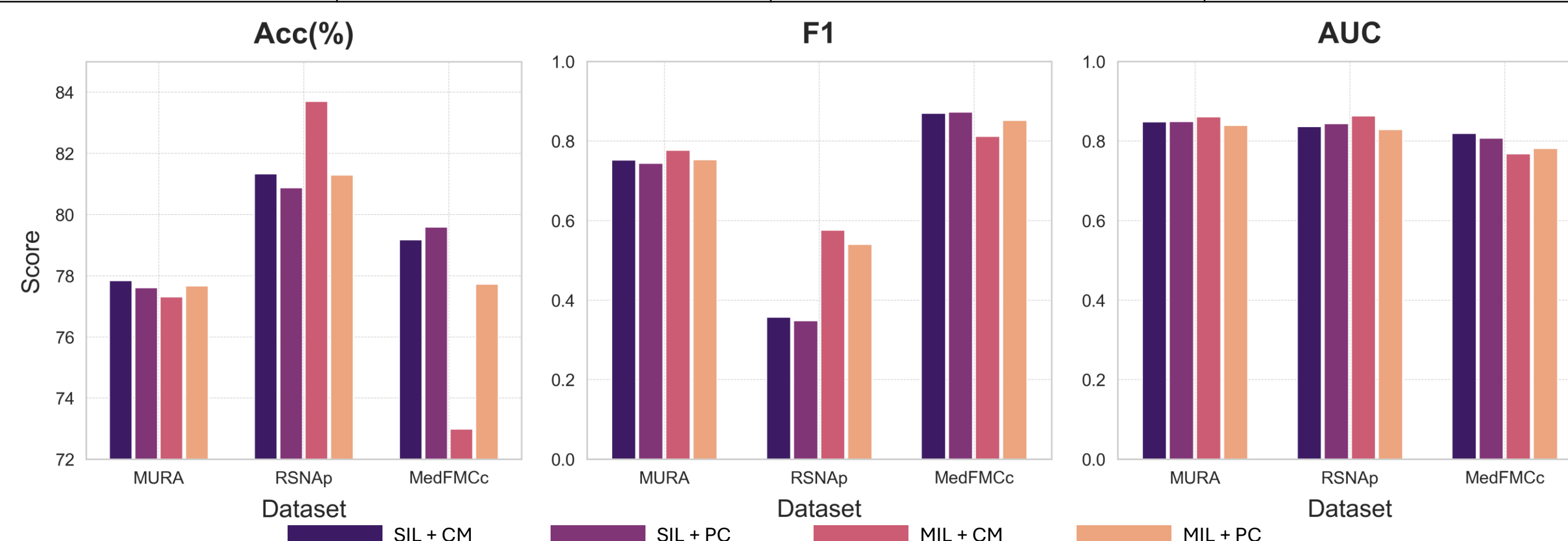
Models	MURA			RSNA _p			MedFMC _c		
	Acc(%)	F1	AUC	Acc(%)	F1	AUC	Acc(%)	F1	AUC
PathRadX (CM + UNI + MIL)	77.31	0.777	0.861	83.70	0.576	0.863	72.99	0.812	0.768
SAM-Med2D	70.94	0.726	0.788	79.72	0.333	0.809	74.23	0.852	0.757
MedCLIP	74.96	0.696	0.844	81.90	0.488	0.834	76.29	0.848	0.784
BiomedCLIP	75.25	0.700	0.835	81.52	0.417	0.820	75.26	0.851	0.762



- PathRadX framework achieved the highest score across all evaluation metrics for the MURA and RSNA_p dataset.
- The proposed method was inferior to other models on the MedFMC_c dataset.
- Overall, PathRadX framework showed competitive results compared to other radiology foundation models.

Comparison of classification performance across strategies.

Strategy				MURA			RSNA _p			MedFMC _c		
SIL	MIL	CM	PC	Acc(%)	F1	AUC	Acc(%)	F1	AUC	Acc(%)	F1	AUC
✓		✓		77.84	0.752	0.848	81.33	0.357	0.836	79.18	0.870	0.819
✓			✓	77.61	0.744	0.849	80.88	0.348	0.844	79.59	0.873	0.807
	✓	✓		77.31	0.777	0.861	83.70	0.576	0.863	72.99	0.812	0.768
	✓		✓	77.67	0.753	0.839	81.30	0.540	0.829	77.73	0.852	0.781



- Utilizing both channel manipulation and MIL showed the highest performance in general, especially on the RSNA_p.
- On the MedFMC_c, equipped with SIL, the proposed method achieved much higher scores than other strategies, even outperforming MedCLIP.

Confusion matrices of evaluation results

		UNI		SAM-Med2D		MedCLIP		BiomedCLIP	
MURA	Predicted Positive	671	221	654	312	486	76	490	75
	Predicted Negative	164	641	181	550	349	786	345	787
	Ground Truth Positive								
	Ground Truth Negative								
RSNA _p	Predicted Positive	296	130	135	75	230	112	176	68
	Predicted Negative	305	1937	466	1992	371	1955	425	1999
	Ground Truth Positive								
	Ground Truth Negative								
MedFMC _c	Predicted Positive	283	54	360	125	321	76	343	103
	Predicted Negative	77	71	0	0	39	49	17	22
	Ground Truth Positive								
	Ground Truth Negative								

- Even if two of the three datasets (RSNA_p and MedFMC_c) were highly imbalanced, the proposed framework demonstrated more balanced and reliable classification performance.

CONCLUSIONS

- In this study, we investigated four different strategies within the PathRadX framework to leverage a pathology foundation model (UNI) on radiology imaging tasks in comparison to radiology foundation models.
- PathRadX proved the hypothesis that pathology foundation models possess sufficient generalization ability so that it can be adapted to radiology images.
- Future work will handle classification performance imbalances among datasets, while apply PathRadX framework to other medical imaging tasks.
- As the proposed framework is confined to 2D medical imaging tasks, we will also explore additional strategies to handle 3D medical images (e.g. lung CT, brain MRI).